PP

Physics-Guided Deep Unfolding Network

Joint Image Restoration & Super-Resolution for Semiconductor Wafer Inspection

Challenge Track: Problem Statement 01 (KLA SEMICON India 2026)

TEAM REGISTRATION DETAILS

Team Name: Pixel Pioneers | Tagline: Restoring reality, one pixel at a time.

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Problem Statement & Semiconductor Inspection Challenges

1. Multiplicative Speckle Noise

- Signal-dependent noise statistics scaling with intensity.
- Pushes pixel intensities outside nominal [0.0, 1.0] bounds.
- Causes thresholding errors in defect metrology.

2. Optical Gaussian Blur

- Light diffraction and point-spread scatter.
- Destroys sub-micron edge sharpness.
- Blurs critical structural boundaries on wafer lines.

3. 2x Spatial Resolution Loss

- Physical optical downsampling (256x256 → 128x128).
- Nanoscale patterns drop below Nyquist sampling limit.
- High-frequency defect geometries are lost.

CORE OBJECTIVE: Jointly denoise, deblur, and super-resolve 2x degraded images back to ground truth fidelity without hallucinating artificial defects on unseen wafer layouts.

Core Idea: Physics-Guided Unrolling vs. Black-Box CNNs

THE BLACK-BOX CNN BOTTLENECK

- Ignores Physical Optics: Treats super-resolution as naive statistical interpolation.
- Hallucinates Non-Existent Patterns: Fabricates fake structural defects on novel layouts.
- Parameter Bloat: Heavy U-Nets / ViTs require 2M to 15M parameters.
- OOD Brittleness: Severe performance drops (>10% SSIM degradation) on novel wafer test samples.
- High Latency: Exceeds edge inspection runtime budgets (>100ms per image).

OUR APPROACH: ALGORITHM UNROLLING (HQS)

- Half-Quadratic Splitting (HQS): Decouples restoration into a physical Data Fidelity step and a Deep Prior step.
- Physical Consistency: Embeds differentiable blur (A) and downsampling (D) operators directly into network graph: $D(A(x)) \approx y$.
- 98% Parameter Reduction: Compact 36.7k parameter footprint (~170 KB weights file).
- Zero Hallucinations: Physics-bounded gradients prevent false pattern formation.
- Real-Time Edge Speed: Achieves 14.95 ms on RTX 5050 and < 3.0 ms on NVIDIA H100.

Proposed Architecture & End-to-End Pipeline Flow

Step 1: Homomorphic Log-VST

v = log(y + 1e-3)

Converts multiplicative speckle into additive Gaussian noise in the log-domain. Stabilizes signal-dependent noise variance before deep learning processing.

Step 2: Scale-Invariant Warm Start

x_0 = GaussianFilter(Bicubic(v, 2x))

Dynamically scales 128x128 input up to 256x256 high-resolution grid. Suppresses high-frequency speckle before unrolled iterations.

Step 3: Unrolled HQS Loop (N=3)

Alternates between:
1. Data Fidelity Block (Physical gradient step on blur/downsample operators)
2. Denoising Prior Block (Shared-weight NAFNet-Lite denoiser).

Step 4: Inverse VST & Calibration

x_clean = exp(z_N) - 1e-3

Applies learned affine calibration: x_out = alpha * x_clean + beta. Final outputs clamped strictly to valid [0.0, 1.0] intensity range.

Innovation & Uniqueness: Core Differentiators

1. Homomorphic Speckle Linearization

Multiplicative speckle noise is transformed via Log-VST. Denoising occurs on stabilized log-intensities, allowing loss functions to optimize structural wafer geometry rather than noise scales.

2. Learnable Optical Operators

The Gaussian blur kernel (sigma, size) and downsampling operators are parameterized and optimized end-to-end via backpropagation, automatically tuning to KLA sensor optics.

3. Activation-Free NAFNet-Lite Denoiser

Replaces non-linear activations (GELU/ReLU) with a Simple Gate (x_1 * x_2) and Simplified Channel Attention. Cuts GPU inference latency by 22% while using only 24 channels.

4. Geometry-Preserving Composite Loss

Multi-scale training objective:
L_total = L1 + 0.5 * L_MS-SSIM + 0.2 * L_SobelEdge + 0.01 * L_KernelReg.
Guarantees sharp edge transitions, structural fidelity, and physically smooth blur kernels.

Quantitative Results: Ours vs. Baseline & Targets

Architecture	Parameters	In-Dist SSIM	In-Dist PSNR	RTX 5050 Latency	H100 Latency (Est.)	OOD Drop	Verification Status
Baseline U-Net	2.1 Million	0.8124	29.20 dB	120.0 ms	~18.0 ms	> 10.0%	PASSED
Physics-DUN (Ours)	**36,710 (~170 KB)**	**0.9115**	**31.84 dB**	**14.95 ms**	**< 3.0 ms**	**< 3.0%**	**PASSED**
KLA Success Target	< 1.0 Million	>= 0.910	>= 31.5 dB	<= 100.0 ms	<= 100.0 ms	Minimal Drop	PASSED

SSIM & PSNR Passed

SSIM 0.9115 (Target >= 0.910)
PSNR 31.84 dB (Target >= 31.5 dB)
High-fidelity sub-micron edge preservation.

Ultra-Light Footprint

36,710 parameters
169.7 KB checkpoint file on disk
Consumes <18% of 1.0 MB budget.

Real-Time Throughput

14.95 ms on laptop RTX 5050 GPU
< 3.0 ms projected on NVIDIA H100
Zero execution bottlenecks.

Technology Stack, Edge Portability & NVIDIA H100 Feasibility

Software Framework & Reproducibility

- Core Framework: PyTorch 2.1.0 (CUDA sm_120 compatibility verified).
- Pinned Dependencies: NumPy 1.24.3, Scikit-Image 0.21.0, Pillow 10.0.0.
- Platform Independent: Fully reproducible via clean pip freeze requirements.txt.

Model Footprint & Memory Efficiency

- Parameter Count: 36,710 parameters (98% smaller than baseline U-Net).
- Checkpoint Size: 169.7 KB on disk (models/best_model.pt).
- VRAM Usage: < 2.8 GB during training; < 500 MB during inference batching.

Inference Profile & H100 Benchmarking

- Laptop GPU (RTX 5050): 14.95 ms per batch of 2 under FP16 mixed precision.
- Datacenter H100 Projection: < 3.0 ms per batch; > 600 images/sec throughput.
- Benchmarking Readiness: Fast I/O batching and tensor transfers optimized.

ONNX Runtime & Edge Deployment

- Portability: Exported and verified to model.onnx (430 KB).
- Structural Integrity: 100% passed onnx.checker validation.
- Cross-Platform: Drop-in deployment for TensorRT, OpenVINO, and C++ runtimes.

Deliverables Package & Verification Checklist

OFFICIAL SUBMISSION REPOSITORY

- Public GitHub Repository URL: https://github.com/ANSHKATIYARAK/semiconductor-hackathon.git

- Public Model Checkpoint Weights (models/best_model.pt): Public Google Drive link pinned in repository README.md (Permissions: Anyone with link can view).

- Submission Pitch Deck: PixelPioneers_KLA_PS01.pdf (8 Slides strictly matching official template).

- i4c Technical Check compliance: Modified entry point to run.py and weights to models/best_model.pt.

VERIFIED ARTIFACTS INVENTORY

[✓] Standalone Evaluation CLI (run.py): Dynamic CLI flags, auto-fallback, zero crash.

[✓] Modular Model Architecture (models/model.py): Complete unrolled HQS pipeline schema.

[✓] Restored Test Predictions (test_predictions/): 400 shape-correct (256x256) .npy files bounded in [0.0, 1.0].

[✓] ONNX Model Export (models/model.onnx): Validated via onnx.checker (430 KB).

[✓] Environment Specification (requirements.txt): Clean, pinned pip freeze output.

[✓] Benchmark Reports: docs/final_report.md and docs/walkthrough.md.